

RAAJVIR VIJAY

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PROFILE

Second-year Industrial Engineering & Management Sciences student at Northwestern and Operations Lead of Northwestern Formula Racing. I work at the intersection of operations research and motorsport — formalising constrained-resource problems, estimating them honestly, and turning them into decisions. Currently running an independent empirical study of Formula 1's aerodynamic development quota, following a data science internship building dynamic pricing and demand models at national retail scale, and earlier research in airline cabin revenue optimisation.

EDUCATION

Northwestern University — Evanston, IL

Expected Class of '29

- BSc Industrial Engineering & Management Sciences, McCormick Engineering | GPA: 3.80
- Coursework: Database Management, Probability, Finance for Engineers

The International School Bangalore — Bengaluru, India

Class of '25

- International Baccalaureate Diploma | Score: 41/45
- IGCSE | 8A* 1A

MOTORSPORT

Operations Lead, Northwestern Formula Racing

2026 – Present

- Run the 120-person build programme across 6 subsystems on a \$50,000 operating budget.
- Restructured project tracking into Jira — milestones, dependencies, visible schedule slip.

f1-atr — Independent Research, F1 Aerodynamic Testing Restrictions

2026

- Built a reproducible Python pipeline over 127 qualifying sessions and 61 team-seasons.
- Validated a lap-time pace metric against championship order at Spearman 0.89–0.98.
- Found the naive design unidentifiable, built a near-tie quasi-experiment over 45 adjacent pairs, and reported the effect as undetectable rather than overstating it.

Co-Host, Six Cylinders F1 Podcast

2024 – Present

- Weekly analysis on Formula 1 economics, commercial strategy and regulation.

EXPERIENCE

Data Science & Consulting Intern, Kaizen Analytix — Atlanta, Georgia

Summer 2026

- Built ThriftIQ, a dynamic pricing algorithm for US thrift retailer Savers'; featured on CNBC.
- Seasonality model clustering 384 stores into 6 climate zones for demand-based pricing.
- Led enterprise adoption of agentic AI tooling across the Data Science department.

Operations Intern, Isvaryam Organic Cooking Oils — Coimbatore, India

Summer 2025

- Applied a Taguchi design-of-experiments model to optimise production cycle timing.
- Presented to leadership; implementation delivered a 16% improvement in seed oil yield.

Salesman, Anaamalais Toyota — Coimbatore, India

Summer 2024

- Supported showroom sales with outreach, test drives and walk-throughs for 200+ customers.
- Managed lead pipeline in CRM; followed up on inquiries and improved conversion tracking.

PROJECTS & RESEARCH

Founder & CEO — Somnia, Sleep-tech Startup (Northwestern Garage)

2023 – Present

- Sleep-tech app with patent-pending biosensor wristband; \$12K pre-seed funding; 1,000+ users.
- TEDx speaker on sleep health; led a 14-person team reaching 1,200+ youth across 5 countries.

Research Author — Airline Cabin Revenue Optimisation, ANA Boeing 787-9

2024

- Modelled optimal seat allocation across business, premium economy and economy on the Tokyo–Los Angeles route using per-class price elasticity of demand and quadratic revenue functions.
- Solved by partial differentiation under a fixed-row constraint, then tested integer configurations; the optimum gained INR 258,000 per flight (2.1%), showing the layout was already near-optimal.

Student Researcher — Alpha Beta Investments, New York

2023

- Built an ML investment risk assessment model in an international team of five.
- OLS model with efficient frontier, CAPM and Monte Carlo simulation; 78% accuracy.

Research Author — Netflix Releases & Stock Price Volatility

2022 – 2023

- Econometric analysis under Prof. Julius Vainora (University of Cambridge), applying OLS regression in Python to test the relationship between content release timing and stock volatility.
- 25-page paper published in the International Journal of Economics and Management Sciences.

Research Assistant, IPR SURA Grant (\$6k) — Northwestern SESP

Summer 2026

- Built Python data models to clean and structure datasets for a five-year behavioural study.

Analyst — Polymarket, via Ascend Consulting Group

Summer 2026

- User research and UX wireframe prototyping; proposed activation-flow improvements.

TECHNICAL SKILLS

- **Tools:** Python (pandas, scikit-learn, statsmodels), MATLAB, Power BI, Excel (VBA), Jira
- **Methods:** OLS and panel regression, fixed effects, quasi-experimental design, power analysis, multiple-comparison correction, constrained optimisation, Monte Carlo simulation, efficient frontier and CAPM, price elasticity of demand, design of experiments (Taguchi), dynamic pricing
- **Software:** Swift, Java, React Native, HTML/CSS/JS
- **Languages:** English, French, Hindi, Tamil, Kannada

AWARDS & RECOGNITION

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| • IIFM Global Economics Olympiad — Rank 1; International Finance Olympiad — Rank 2 | 2024 |
| • Blue Ocean Entrepreneurship Challenge — 3rd in Asia + Oceania (Top 10 global) | 2024 |
| • UBER Global Hackathon — Regional Finalist | 2022 |

REFERENCES

Andreas Puppa — Supervisor — VP, Data Science, Kaizen Analytix — andreas.p@kaizenanalytix.com

Prof. David Gatchell — Professor — Director, Manufacturing & Design Eng. — d-gatchell@northwestern.edu

Prof. Julius Vainora — Research Supervisor — Asst Professor of Economics. — jv429@cam.ac.uk